GRADUATE TREATISE & COMPUTATIONAL LAB

QUANTA

An Epistemological Autopsy of Quantum Ruptures: Analytical Mechanics to Non-Locality (1895–1980s)

> *"Die gegenwärtige Lage der theoretischen Physik ist derart, daß man es wohl für aussichtslos halten muß, auf dem Wege kontinuierlicher Theorien zu einer befriedigenden Auffassung der Erscheinungen zu gelangen."*
> — Max Planck, Vorlesungen über die Theorie der Wärmestrahlung (1906)

CHAPTER I 1895–1905

Thermodynamic Inconsistencies & The Quantum of Action

The ultraviolet catastrophe of classical equipartition and Planck's thermodynamic interpolation of cavity resonator entropy derivatives $\frac{d^2S}{dU^2}$.

1.1 The Epistemic Crisis of Classical Equipartition

In classical thermodynamics and electrodynamics, the Rayleigh-Jeans spectral density emerges unconditionally from the equipartition theorem applied to the infinite modes of an electromagnetic cavity of volume V:

As frequency $\nu \to \infty$, the total electromagnetic energy density diverges catastrophically:

Conversely, Wien's 1896 phenomenological empirical law, $u_W(\nu, T) = \alpha\nu^3 \exp(-\beta\nu/T)$, matched high-frequency luminescence data but failed decisively in the far-infrared Rubens-Kurlbaum experiments (1900).

MAX PLANCK'S THERMODYNAMIC DERIVATION (OCTOBER 1900)

Planck analyzed the second derivative of the entropy S of a virtual monochromatic Hertzian dipole oscillator of equilibrium internal energy U:

Planck constructed the simplest rational mathematical interpolation uniting both empirical asymptotes:

Integrating twice with boundary conditions $\frac{dS}{dU} = \frac{1}{T}$ yields the exact spectral radiance and oscillator energy:

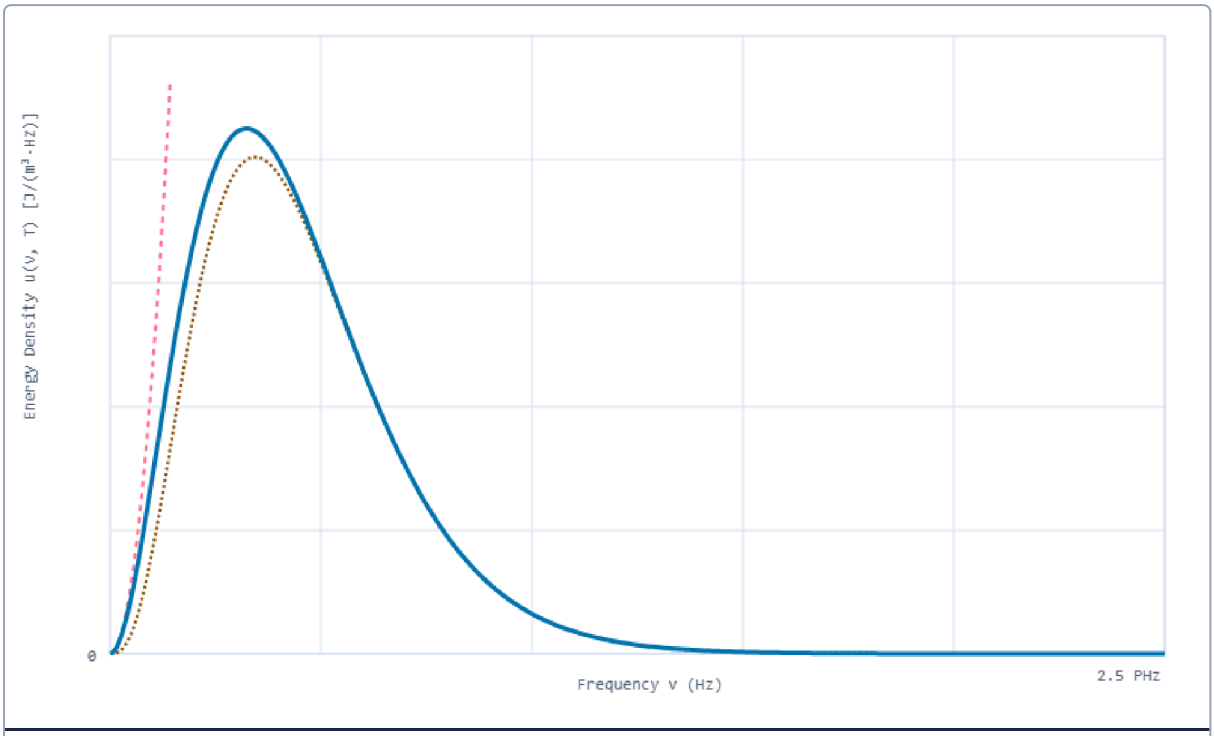
1.2 Einstein's 1909 Fluctuation Inversion

Albert Einstein inverted Boltzmann's entropy formula $S = k_B \ln W$ to calculate the mean square energy fluctuations $\langle (\Delta E)^2 \rangle$ in a sub-volume V of blackbody radiation:

This proved incontrovertibly that radiation simultaneously exhibits discrete particle-like quantum fluctuations and continuous classical wave interference.

SPECTRAL RADIANCE & FLUCTUATION ENGINE

NUMERICAL SOLVER



Planck Law Wien Law Rayleigh-Jeans

WIEN PEAK FREQUENCY (N_{MAX})

3.23 × 10¹⁴ Hz

$\nu_{\max} = 2.821\, k_B T/h$

PEAK RADIANCE $U(N_{\mathrm{MAX}})$

1.32 × 10⁻¹⁵ J/m³·Hz

Stefan-Boltzmann scaling

EINSTEIN FLUCTUATION INVERSION $\langle(\Delta E)^2\rangle$

Particle Noise: 94.0% | Wave Noise: 6.0%

Calculated for $V = 1.0 \times 10^{-6}\ \mathrm{m}^3$ at $\nu_{\max}$

CHAPTER II 1913–1925

Dispersion Theory & The Birth of Matrix Mechanics

The breakdown of classical Fourier kinematics in Bohr-Sommerfeld orbits, Kramers-Heisenberg dispersion, and Heisenberg's revolutionary *Umdeutung* coordinate algebra $[X, P] = i\hbar I$.

2.1 The Failure of Classical Fourier Kinematics

In the Old Quantum Theory (Bohr-Sommerfeld-Epstein), electrons revolved in invariant Keplerian orbits quantized by the action integrals:

Classical electrodynamics dictated that the orbital motion generates radiation at orbital frequency ω_{orb} and its integer harmonics $m\,\omega_{orb}$. But experimentally, atoms emitted light at **Ritz transition frequencies**:

This fundamental discrepancy led Hendrik Kramers and Werner Heisenberg (1924–1925) to replace the unobservable continuous trajectory $x(t)$ with a two-index quantum amplitude array $X_{mn}e^{i\omega_{mn}t}$.

KRAMERS-HEISENBERG DISPERSION & THOMAS-REICHE-KUHN SUM RULE

By studying atomic dipole scattering of electromagnetic radiation of frequency ν, Kramers and Heisenberg derived the quantum polarizability tensor:

The Thomas-Reiche-Kuhn (TRK) sum rule imposes an exact conservation constraint on oscillator strengths f_{kn}:

In June 1925 on Helgoland, Heisenberg realized that multiplying two quantum observable arrays X and Y requires a non-commutative convolution rule:

2.2 The Born-Jordan-Heisenberg Drei-Männer-Arbeit (1925)

Max Born recognized Heisenberg's multiplication law as Cayley's matrix algebra. Born and Pascual Jordan proved the fundamental canonical commutation relation:

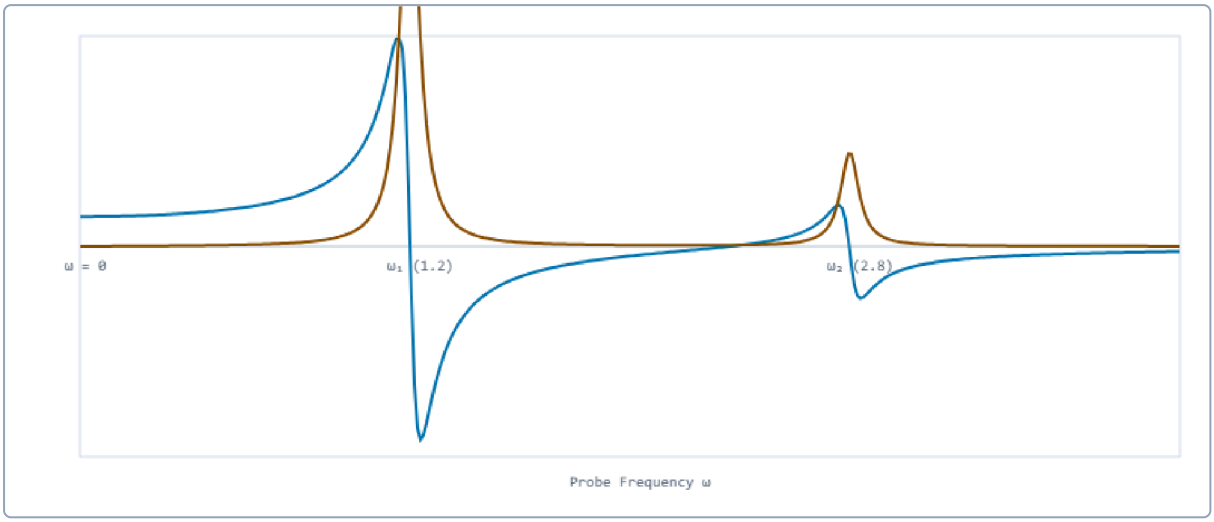
Taking the trace of both sides over any finite $N \times N$ dimensional matrix space reveals:

\text{Tr}([X, P]) = \text{Tr}(XP - PX) = \text{Tr}(XP) - \text{Tr}(XP) = 0 \neq i\hbar N">

Hence, quantum kinematic observables **must be infinite-dimensional unbounded operators**.

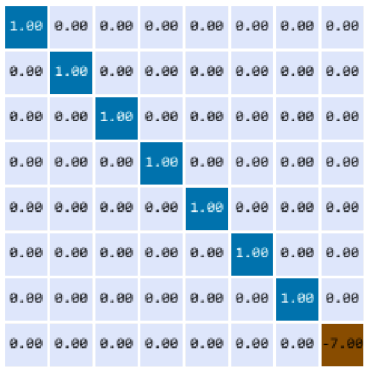
DISPERSION & MATRIX COMMUTATOR SOLVER

MATRIX ALGEBRA



Transition $n \to k$	Energy $\hbar\omega_{kn}$	Dipole $\lvert x_{kn}\rvert^2$	Oscillator f_{kn}
$\lvert 1\rangle \to \lvert 2\rangle$ (Resonance $\omega_1 = 1.2$)	1.200 $\hbar$	0.2708	0.6500
$\lvert 1\rangle \to \lvert 3\rangle$ (Resonance $\omega_2 = 2.8$)	2.800 $\hbar$	0.0625	0.3500
Total Sum $\sum f_{kn}$ (TRK Rule):			1.0000

TRUNCATED COMMUTATOR $[X_N, P_N]$ HEATMAP ($N = 8$)



COMMUTATOR DEVIATION FROM CANONICAL IDENTITY

‖[X_N, P_N] - iħ I‖_F = 8.0000 (Boundary Truncation Defect at N=8)

Vanishes asymptotically as $N \to \infty$ for all interior matrix states

CHAPTER III 1923–1926

Hamilton-Jacobi Geometry & Wave Mechanics

The optical-mechanical analogy of Hamilton, Louis de Broglie's phase waves, and Erwin Schrödinger's variational derivation of the wave equation via action variables $S = \hbar \arg \psi$.

3.1 Hamilton's Optical-Mechanical Analogy

In 1834, William Rowan Hamilton revealed that classical trajectory mechanics and geometrical wave optics share identical mathematical topologies. Fermat's principle of least time in optics corresponds exactly to Maupertuis' principle of least action in dynamics:

The refractive index $n(x)$ corresponds to the local classical momentum $p(x) = \sqrt{2m(E - V(x))}$. Hamilton observed that just as geometric optics is the short-wavelength limit ($\lambda \to 0$) of wave optics, Newtonian mechanics must be the short-wavelength limit of an underlying wave physics.

SCHRÖDINGER'S 1926 VARIATIONAL DERIVATION

Schrödinger substituted Hamilton's principal function $S(x,t) = W(x) - Et$ into the stationary Hamilton-Jacobi equation:

Instead of treating S directly, Schrödinger introduced the logarithmic ansatz $S = K \ln \psi$, choosing $K = \hbar$:

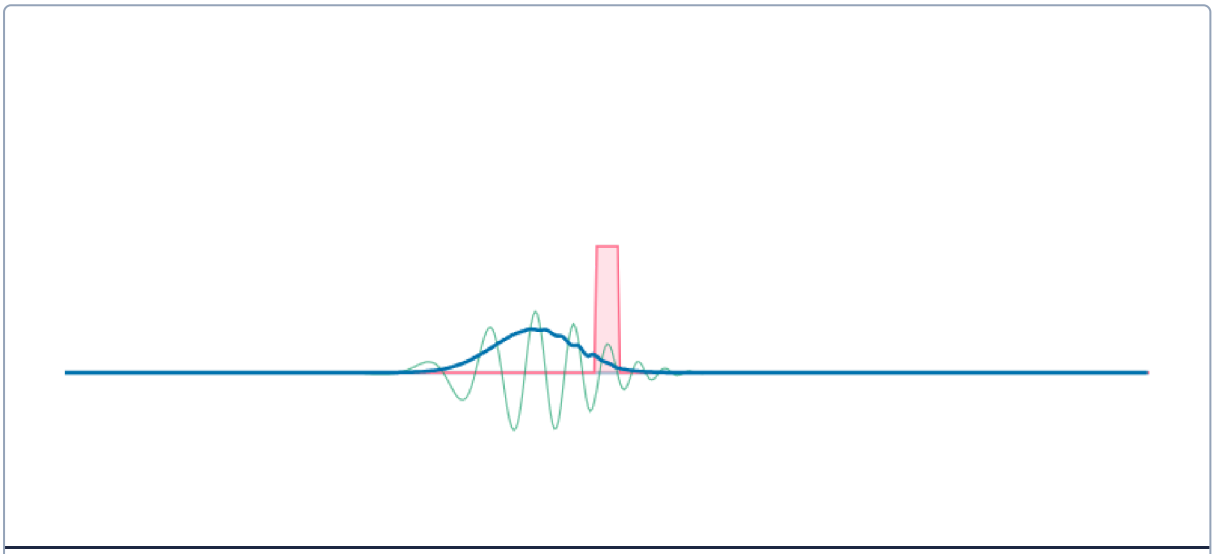
Constructing the action functional $\mathcal{J}[\psi] = \int \left[\frac{\hbar^2}{2m} |\nabla \psi|^2 + (V(x) - E)|\psi|^2 \right] d^3x$ and asserting Euler-Lagrange stationarity $\delta \mathcal{J} = 0$ yields the celebrated time-dependent wave equation:

3.2 Split-Step Fourier Numerical Evolution

To evolve $|\psi(x,t)|^2$ with machine precision and unconditional unitarity, we employ the Strang operator splitting:

This exact unitary transformation guarantees norm conservation $\int |\psi|^2 dx \equiv 1$ to numerical roundoff ($\sim 10^{-14}$).

SPLIT-STEP SCHRÖDINGER EVOLVER REAL-TIME PDE SOLVER (512 MESH)



$|\psi(x)|^2$ Density $\mathrm{Re}\{\psi(x)\}$ Potential $V(x)$



UNITARITY NORM $\int |\Psi|^2 DX$

1.00000000

Preserved to machine precision

TUNNELING TRANSMISSION T

1.6%

WKB barrier penetration

CHAPTER IV 1926–1932

Transformation Theory & Mathematical Axiomatics

The unification of wave and matrix mechanics by Paul Dirac and Pascual Jordan, unbounded self-adjoint operators, and John von Neumann's rigorous Hilbert space formalization.

4.1 Dirac-Jordan Transformation Calculus

In 1927, Paul Dirac and Pascual Jordan independently demonstrated that Schrödinger's wavefunctions $\psi(x)$ and Heisenberg's matrices X_{mn} are simply distinct coordinate representations of abstract state vectors $|\psi\rangle$ living in an infinite-dimensional Hilbert space $\mathcal{H}$:

Any two complete orthonormal bases $\{|a\rangle\}$ and $\{|b\rangle\}$ are linked by a unitary transformation operator $\hat{U}$:

VON NEUMANN'S RIGOROUS FOUNDATIONS (1932)

Dirac's delta function $\delta(x)$ and continuous bras $\langle x|$ lacked rigorous mathematical grounding under classical measure theory. John von Neumann provided the definitive axiomatic formulation:

1. States correspond to one-dimensional rays in a complex separable Hilbert space $\mathcal{H}$.
2. Physical observables are **densely defined, unbounded self-adjoint operators** ($\hat{A} = \hat{A}^*$, where domains $\mathcal{D}(\hat{A}) = \mathcal{D}(\hat{A}^*)$).
3. By the **Spectral Theorem**, any self-adjoint operator $\hat{A}$ admits a unique projection-valued measure $E(\lambda)$:

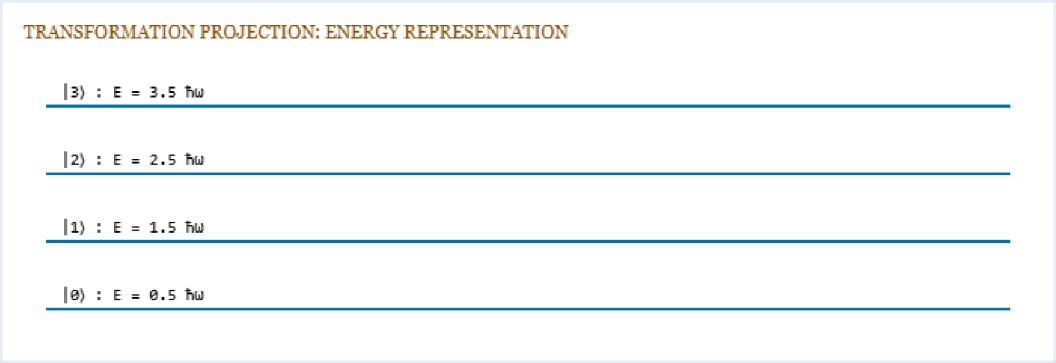
4.2 The Stone-von Neumann Theorem

Let $(\hat{U}(s), \hat{V}(t))$ be strongly continuous one-parameter unitary groups satisfying the Weyl commutation relation:

The Stone-von Neumann theorem proves that for finite degrees of freedom, any irreducible representation of this algebra is **unitarily equivalent** to the Schrödinger position-momentum representation. Wave and matrix mechanics are mathematically isomorphic.

SPECTRAL DECOMPOSITION & BASIS ROTATOR

HILBERT SPACE VISUALIZER



MATRIX ELEMENTS ⟨n | Ĥ | m⟩ (DIAGONAL ENERGY BASIS)

E_0 **0.5 ħω**	⟨0|H|1⟩ 0.000	⟨0|H|2⟩ 0.000	⟨0|H|3⟩ 0.000
⟨1|H|0⟩ 0.000	**E_1** **1.5 ħω**	⟨1|H|2⟩ 0.000	⟨1|H|3⟩ 0.000
⟨2|H|0⟩ 0.000	⟨2|H|1⟩ 0.000	**E_2** **2.5 ħω**	⟨2|H|3⟩ 0.000
⟨3|H|0⟩ 0.000	⟨3|H|1⟩ 0.000	⟨3|H|2⟩ 0.000	**E_3** **3.5 ħω**

UNITARY INVARIANCE RESIDUAL $\|U^\dagger U - I\|$

0.00 × 10⁻¹⁶

Spectral projection completeness verified: $\sum |k\rangle\langle k| = \hat{I}$

CHAPTER V 1935–1967

Measurement, Non-Locality & Realism Theorems

The Einstein-Podolsky-Rosen reality criterion, Bohm's spin singlet reformulation, John Stewart Bell's inequality, and the experimental CHSH violation.

5.1 The 1935 EPR Paradox & Local Realism

Einstein, Podolsky, and Rosen asserted that an adequate physical theory must satisfy **Local Realism**:

"If, without in any way disturbing a system, we can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to this quantity."
For an entangled singlet state $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, measuring particle A instantly determines particle B's spin with unit certainty across arbitrary spacelike separations. EPR concluded that quantum mechanics must be incomplete, missing "local hidden variables" λ.

BELL'S THEOREM & THE CHSH INEQUALITY (1964, 1969)

John Stewart Bell demonstrated that *any* theory founded on local hidden variables $\lambda \in \Lambda$ satisfying:

must obey the Clauser-Horne-Shimony-Holt (CHSH) bound on correlation expectations:

However, quantum mechanics predicts the correlation for a singlet state:

Choosing the optimal configuration $\theta_a = 0, \theta_{a'} = \frac{\pi}{2}, \theta_b = \frac{\pi}{4}, \theta_{b'} = \frac{3\pi}{4}$ yields:

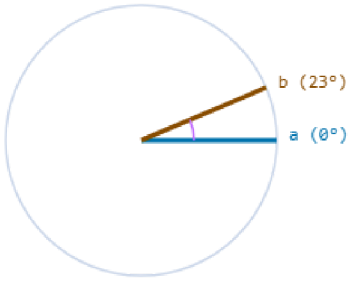
This reaches the ultimate **Tsirelson Bound** $2\sqrt{2}$, decisively refuting local realism in Nature.

5.2 Kochen-Specker Contextuality (1967)

In Hilbert spaces of dimension $\dim \mathcal{H} \geq 3$, quantum observables cannot be assigned pre-existing truth values $\{0, 1\}$ independent of the measurement context (commuting co-measured operators). Reality is non-contextually irreconcilable.

BELL-CHSH ENTANGLEMENT ENGINE

MONTE CARLO LAB (10,000 PAIRS)



CHSH CORRELATION PARAMETER (S)

2.394

Local Realist Bound: $S \leq 2.000$

INEQUALITY VIOLATION STATUS

VIOLATED (+3.9 σ)

Tsirelson Bound: $2\sqrt{2} \approx 2.828$

INDIVIDUAL PAIR CORRELATIONS

E(a,b) = -0.921 | E(a,b') = -0.375 | E(a',b) = -0.927 | E(a',b') = -0.921

Monte Carlo sample variance $\sigma < 0.008$

CHAPTER VI 1948–1980s

Path Integrals, Decoherence & Einselection

Richard Feynman's sum over histories, the classical limit via stationary phase $\delta S = 0$, and Wojciech Zurek's environment-induced superselection (einselection).

6.1 Feynman's Functional Path Integral

In 1948, Richard Feynman revolutionized quantum dynamics by formulating the quantum transition amplitude $K(x_b, t_b; x_a, t_a)$ as a functional integral over all continuous paths $x(t)$:

where $S[x(t)] = \int_{t_a}^{t_b} \mathcal{L}(x, \dot{x}, t)\, dt$ is the classical action.

In the macroscopic correspondence limit ($\hbar \to 0$ or $S \gg \hbar$), the phase $\frac{S}{\hbar}$ oscillates with violent frequency across non-classical paths, completely canceling via destructive interference. The only surviving trajectory is the stationary point of the action:

ENVIRONMENT-INDUCED SUPERSELECTION & DECOHERENCE

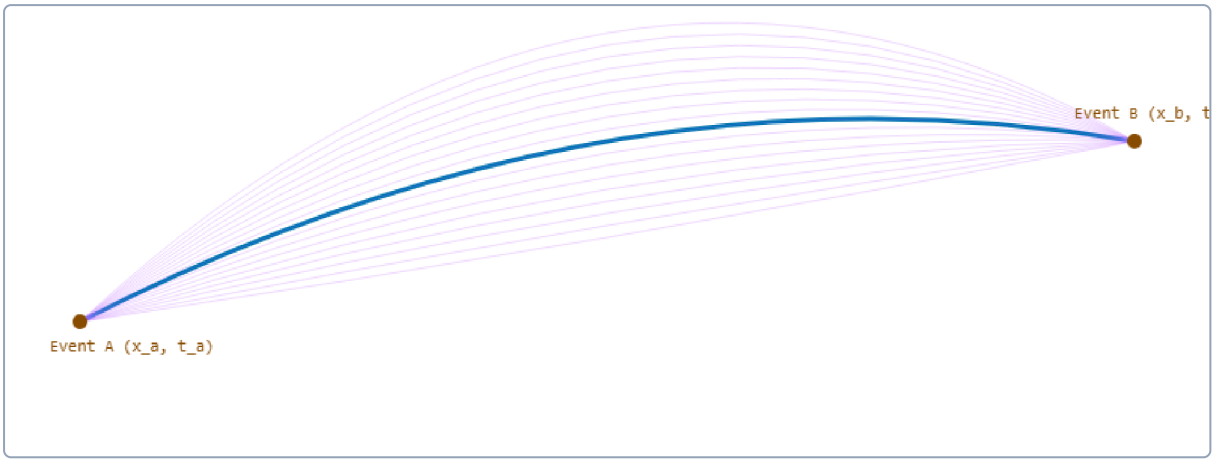
The measurement problem is resolved without ad-hoc wavepacket collapse via open quantum systems theory (Zeh 1970, Zurek 1981). A central system $\mathcal{S}$ couples to an environmental bath $\mathcal{E}$ via Hamiltonian $\hat{H} = \hat{H}_S + \hat{H}_E + \hat{H}_{int}$.

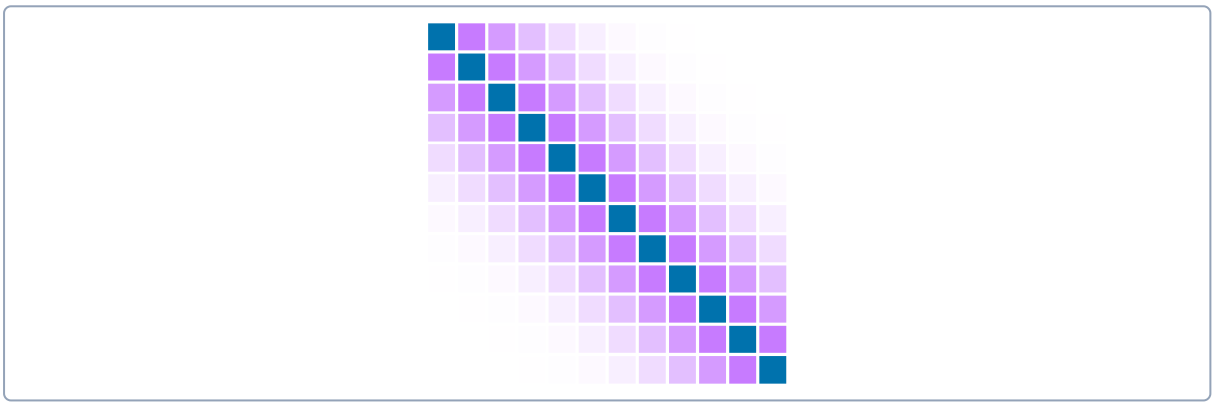
Tracing out the unobserved reservoir degrees of freedom leaves the **reduced density matrix** $\hat{\rho}_S = \mathrm{Tr}_E(\hat{\rho}_{total})$ governed by the Caldeira-Leggett master equation:

The off-diagonal coherence terms decay exponentially at the decoherence rate:

PATH INTEGRAL & DECOHERENCE ENGINE

FEYNMAN TRAJECTORIES & DENSITY MATRIX





STATE PURITY $\Gamma = \mathrm{TR}(P^2)$

0.743

Pure State: 1.00 | Mixed: $\to 0.50$

DECOHERENCE TIMESCALE T_D

1.24 × 10⁻¹² s

Pointer basis einselected